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Revolutionizing Well Data Access and Analysis with AI: A Case Study and Insights Major Middle Eastern Oil Fields

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Abstract

In the quest for operational excellence, petro-technical experts require comprehensive visibility into the myriad data and events spanning the lifecycle of each well. This task is daunting, given that crucial data and expertise are dispersed across numerous databases, thousands of documents, and among hundreds of experts, many of whom are retired or nearing retirement. To address this challenge, operator has developed an integrated digital knowledge layer for wells, aimed at empowering users to easily access information, events, and knowledge related to operating wells. This facilitates periodic 360-degree well reviews to ensure wells are performing as expected and to identify opportunities for performance enhancement.

The AI Digital Well Book, an in-house solution, offers a unified view of diverse data elements across various disciplines. Each well and its associated data elements are organized into chapters, capturing key events, gathered data, analyzed data, and live data, represented as individual pages. AI Digital Well Book is architected around three synergistic layers.

- Data Integration and Synthesis Layer (Foundation Layer): It includes key functions such as data ingestion, semantic normalization, event extraction and knowledge graph linking entities
- AI and Analytics Layer (Intelligence Layer): This layer applies advanced analytics and AI models to derive insights, detect anomalies, and support decision-making.
- Interaction layer: This top layer provides an intuitive, conversational and Visualization interface for engineers and domain experts to interact with the system.

Digital Well book allows users to visualize all events on a timeline, irrespective of data storage locations. The system extracts events from data, presents them within the well lifecycle, and connects these events through common oil and gas entities and concepts. It organizes and indexes data around well concepts such as equipment, vendors, well activities, problems, maintenance history, intervention events, reservoir monitoring, rig-less performance analysis, drilling issues, reservoir and petroleum engineering analysis, health and safety events, and incidents.

The solution supports well and reservoir management review processes in a standardized, structured, and centralized manner, significantly enhancing knowledge sharing between assets by documenting best

practices and lessons learned across 5000+ wells. Engineers have experienced a 60% reduction in time spent searching for data during well reviews. Additionally, the solution enables easy and effective searches of historical information, facilitating fact-based, informed decisions regarding well abandonments, interventions, and planning efforts. It provides a single source of data, faster access, improved quality checks, and integrated workflows, thereby simplifying collaboration.

A notable feature of this solution is Generative AI real time data agent, an interactive chat-based tool that allows petroleum and reservoir engineers to interact with well data contextually. This agent highlights issues related to wells, enhancing the engineers' ability to manage and optimize well performance.

Introduction: The Imperative for Digital Well Knowledge Management

The upstream oil and gas sector faces an unprecedented challenge: managing the vast, complex, and ever-growing volume of data generated throughout the lifecycle of thousands of wells while simultaneously confronting a significant loss of experienced personnel through retirement. Crucial knowledge about well performance, interventions, geoscience interpretations, and operational lessons learned often remains trapped in disparate databases, unstructured documents, and the minds of experts. This fragmentation severely hinders operators' ability to achieve operational excellence, optimize production, ensure safety, and make timely, informed decisions. Traditional methods of data retrieval and review are time-consuming, error-prone, and fail to provide the holistic, contextual understanding required for effective well and reservoir management (WRM). Operating a vast portfolio of over thousands of wells, recognized this critical gap and embarked on a transformative journey to build the "AI Digital Well Book" – an integrated digital knowledge layer leveraging semantic technology, AI, and Generative AI to empower its petro-technical community.

Current State

The "as-is" state of well data and knowledge management in largest operators, prior to this initiative, is characterized by several key challenges:

- **Data Silos:** Well data resides across numerous specialized systems – drilling reporting tools, production databases (PI/OSIsoft), well integrity systems, SCADA, completion databases, reservoir simulation models, subsurface interpretation platforms, document management systems and countless spreadsheets and reports. Integration is often point-to-point, brittle, and incomplete.
- **Unstructured Information Overload:** A significant portion of critical knowledge (lessons learned, operational reports, emails, meeting notes, expert analyses) exists in unstructured text documents (PDFs, Word, emails), making it virtually invisible to traditional database queries.
- **Inefficient Knowledge Retrieval:** Engineers spend an estimated 30-60% of their time searching for and validating data across these disparate sources, significantly delaying reviews and decision-making (consistent with industry surveys from SPE, DNV, and McKinsey).
- **Contextual Disconnect:** Events, analyses, and data points exist in isolation. Understanding the sequence of interventions, their impact on production, related integrity issues, and underlying geological factors requires manual correlation across systems, a process prone to oversight.
- **Expertise Atrophy:** As senior petro-technical staff retire, invaluable tacit knowledge about specific wells, reservoirs, historical issues, and successful (or unsuccessful) strategies risks being lost forever if not systematically captured and contextualized.
- **Standardization Deficits:** Naming conventions, units, and data quality vary significantly across assets, time periods, and source systems, hindering automated analysis and comparison.

- **Reactive vs. Proactive:** Reviews are often triggered by problems or planned interventions, lacking a continuous, holistic view of well performance and health, potentially missing early warning signs or optimization opportunities.

All major operators have acknowledged similar challenges and are investing heavily in data foundations and analytics to overcome them. However, achieving a truly integrated, contextualized, and easily accessible "well story" remains a significant hurdle.

GAPs in current processes

The current state creates critical gaps impacting safety, efficiency, and value:

- **Suboptimal Decision Making:** Decisions regarding interventions, workovers, abandonment, or production optimization are often made without a complete historical and contextual picture, increasing risk and potentially leaving value unrealized.
- **Repeated Mistakes:** Lessons learned from past failures or inefficiencies are not easily discoverable, leading to the repetition of costly errors.
- **Ineffective Well Reviews:** Periodic well reviews (WRM, integrity) are hampered by incomplete data, lengthy preparation times, and difficulty identifying trends or correlations across disciplines.
- **Slow Problem Resolution:** Diagnosing well problems requires piecing together information from multiple sources, delaying corrective actions and impacting production.
- **High Onboarding Costs:** New engineers face a steep learning curve to understand the history and context of specific wells and fields.
- **Compliance & Audit Risks:** Demonstrating compliance with regulations or internal standards is difficult when evidence is scattered and hard to consolidate.
- **Missed Optimization Opportunities:** Subtle trends or correlations indicating underperformance or enhancement potential are easily overlooked in fragmented data landscapes.
- **Increased Maintenance Overhead:** Managing multiple dashboards across departments increases IT complexity and support costs.

Future state: AI and Generative AI as catalysts for Transformation

The envisioned future state, exemplified by AI Digital Well Book, leverages advanced technologies to bridge these gaps:

- **Unified Contextual View:** A single, holistic digital representation of each well, integrating structured and unstructured data, events, and analyses throughout its lifecycle.
- **Proactive Insights & Anomaly Detection:** AI models continuously analyze integrated data streams to identify early signs of performance degradation, integrity issues, or optimization potential, shifting from reactive to proactive management.
- **Intelligent Search & Knowledge Retrieval:** Semantic search and Generative AI enable engineers to ask complex, contextual questions in natural language ("Show me all wells in Reservoir X with ESP failures within 6 months of acid stimulation, along with the stimulation reports and subsequent production trends") and receive synthesized, relevant answers with source references.
- **Automated Event Correlation:** Systems automatically link related events (e.g., a drilling incident, a subsequent completion modification, and its long-term production impact) based on shared entities and timelines.
- **Preserved & Augmented Expertise:** Tacit knowledge is systematically captured, contextualized within the well's digital twin, and made accessible. Generative AI agents act as "digital assistants," augmenting human expertise by providing instant access to relevant historical data and analyses.

- **Streamlined & Standardized Workflows:** Integrated platforms enforce data standards and provide seamless access within core workflows (e.g., well review, intervention planning, abandonment studies), improving efficiency and consistency.
- **Enhanced Collaboration:** A centralized, trusted source of truth facilitates cross-disciplinary collaboration and knowledge sharing across different assets and teams.

Technical and digital Architecture: The Three-layer Foundation

The AI Digital Well Book is architected for scalability, flexibility, and intelligence:

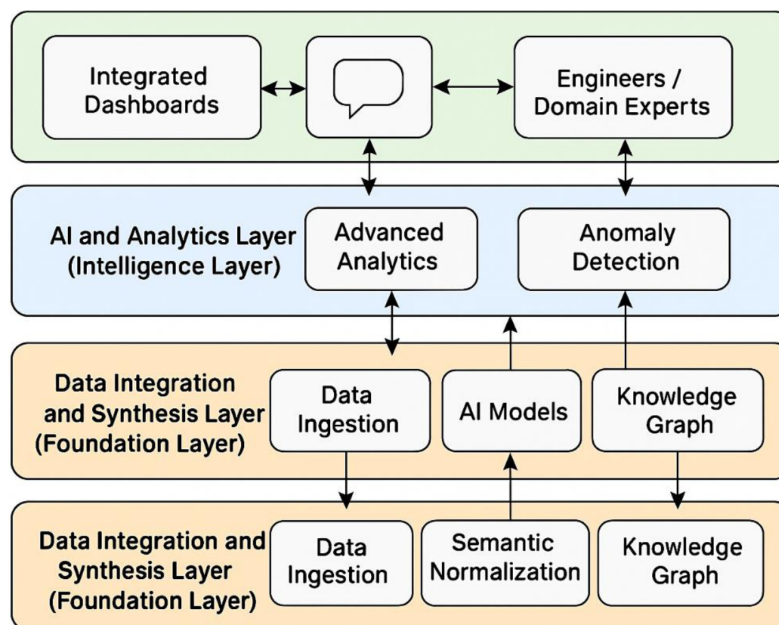


Figure 1

Layer 1: Data Integration and Synthesis (Foundation Layer)

This foundational layer is responsible for ingesting, harmonizing, and contextualizing data from various sources:

- **Data Ingestion:** Robust pipelines connect to diverse sources: real-time sensors (PI), drilling/completion databases (OpenWells, drilling & work-over schedules), production allocation systems, reservoir engineering tools (Petrel, Eclipse), document repositories (SharePoint), maintenance systems (SAP), GIS, and legacy databases. APIs, ETL/ELT processes, and change data capture are employed.
- **Semantic Normalization:** A critical step. Data from different sources is mapped to a unified, domain-specific ontology. This resolves naming conflicts (e.g., "Well_ID" vs. "UWI"), standardizes units, and defines relationships.
- **Event Extraction:** NLP techniques automatically extract key events (spud date, completion, workover, failure, test, incident) from unstructured text reports, emails, and logs, structuring them with timestamps and relevant entities.
- **Knowledge Graph Construction:** The core organizing principle. Extracted entities (Wells, Equipment, Formations, Activities, Problems, Chemicals, Vendors, Personnel, Documents) and their relationships (Well has Equipment, Activity caused Problem, Document describes Event) are stored in a graph database (e.g., Neo4j, AWS Neptune, Azure Cosmos DB). This creates a dynamic, interconnected model of the well universe. This is currently work in progress as part of the solution.

Layer 2: AI and Analytics (Intelligence Layer)

This layer leverages advanced analytics and AI models to deliver powerful insights, detect anomalies, and enhance decision-making across the well lifecycle. Descriptive analytics provide dashboards and visualizations that summarize well status, performance history, integrity metrics, and key event timelines. Diagnostic analytics enable root cause analysis by correlating events such as specific workover types with subsequent failure rates or identifying geological factors influencing sand production. Predictive analytics apply machine learning to forecast equipment failures (such as ESPs and downhole gauges), production decline, and intervention success probabilities. Prescriptive analytics go a step further by recommending optimal actions, such as the best intervention type, optimal choke settings, or forecast-driven lift gas allocation. Complementing these capabilities, anomaly detection identifies deviations from expected behavior in real-time sensor data or aggregated performance metrics, while a generative AI agent provides contextual, conversational access to well data, allowing engineers to ask questions and receive intelligent, actionable responses.

Layer 3: Interaction Layer (User Experience)

The top layer provides an intuitive interface for engineers and domain experts to seamlessly interact with the system. A visualization hub offers interactive timelines of well events, cross-plots linking different data types such as production, downhole pressure, and injection, along with geospatial views and schematic visualizations of wellbore equipment history. A key enhancement in this layer is the introduction of centralized dashboards, which consolidate KPIs, operational events, and analytical insights into a single unified interface. Unlike traditional distributed dashboards developed independently by different departments or assets, these centralized dashboards ensure consistency, accuracy, and real-time visibility across the organization. The flagship feature of this layer is the conversational interface powered by a custom, fit-for-purpose LLM, which understands complex, context-rich queries in natural language. It retrieves relevant data from the Knowledge Graph and underlying sources, synthesizes answers, provides summaries, highlights potential issues

Knowledge graph

The Knowledge Graph (KG) is the semantic backbone enabling the solution's intelligence. Knowledge graph is a structured, flexible data model representing entities (nodes) and their relationships (edges). It moves beyond rigid tables to capture the rich, interconnected nature of well data. This is currently work in progress as part of overall architecture.

- **Entity Resolution:** Links data about the same real-world thing (e.g., "Well-123," "ESP Pump SN#789," "X formation") from different source systems into a single node.
- **Contextual Linking:** Explicitly defines relationships: Well-123 - HAS_COMPLETION -> Completion-2020-01 - USED -> Sliding Sleeve Manufacturer-X - ASSOCIATED_WITH -> Report-FailureAnalysis-2022.pdf. Workover-2022-05 - ADDRESSED -> Problem-ESP-Failure - OCCURRED_IN -> Well-123.
- **Semantic Search:** Enables complex queries traversing relationships: "Find completion reports for wells completed in the X Formation that experienced casing corrosion within 5 years, where the cement vendor was Company-Y."
- **Foundation for AI:** Provides the structured context crucial for training accurate ML models and enabling the Generative AI agent to understand domain concepts and relationships when answering questions or generating insights.
- **Linking Business Entities:** The inherently links technical data to business context:
 - Equipment <-> Vendors <-> Maintenance Costs <-> Reliability Metrics

- Well Activities <-> Service Companies <-> Performance KPIs <-> HSE Incidents
- Production Problems <-> Interventions <-> Cost <-> Production Uplift <-> NPV
- This enables holistic analysis of technical performance alongside cost, reliability, and safety impact

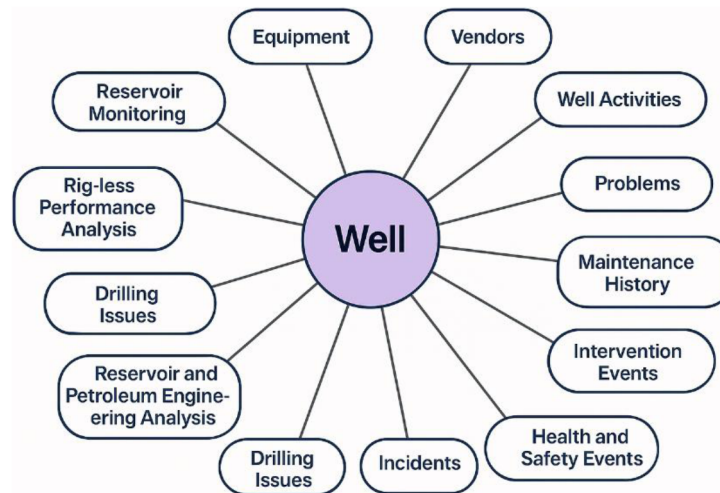


Figure 2

Case Study: Implementation Approach: Phased, Agile, and User-Centric

The adoption of a pragmatic and scalable implementation strategy to ensure the successful deployment of the AI Digital Well Book. The approach was structured around clear business objectives, agile development principles, and continuous user engagement. The following

Vision and Scope Definition

The initiative was launched with a clearly defined vision centered around three primary objectives: reducing the time spent searching for well data, enhancing the quality and consistency of well reviews, and capturing and preserving expert knowledge. The scope of the project was deliberately focused to ensure effective implementation, beginning with a targeted set of assets for initial deployment. It also included the integration of core data sources such as production data, intervention records, and document repositories. Emphasis was placed on key well concepts including activities, equipment, and performance metrics. To measure the initiative's impact, success metrics were established from the outset, covering areas such as time savings, user adoption rates, and improvements in the review cycle

Pilot and Proof of Value

A pilot phase was initiated to validate the solution using a carefully selected subset of around 1,000 wells, focusing on high-impact use cases to ensure meaningful results and strong user engagement. This phase successfully demonstrated the technical feasibility of the architecture, delivered tangible value to end users, and provided critical feedback for further refinement. Key focus areas included preparing for periodic well reviews, retrieving historical intervention data, and identifying performance anomalies, all of which showcased the solution's ability to enhance efficiency and decision-making.

Agile development and phased roll-out

The solution was developed using agile methodologies—Scrum and Kanban—to support rapid iteration, continuous delivery, and responsiveness to evolving user needs. The rollout was strategically divided into four progressive phases, each designed to build technical maturity and deliver increasing business value.

1. Phase 1: Core Data Integration
 - Ingestion of key operational and historical data sources
 - Construction of a foundational knowledge graph linking wells, activities, and documents
 - Deployment of timeline visualizations to contextualize well events
2. Phase 2: Advanced Knowledge Graph and Analytics
 - Expansion of the graph to include equipment, operational problems, and analytical relationships
 - Introduction of semantic search for intuitive data discovery
 - Deployment of descriptive and diagnostic analytics to support decision-making
3. Phase 3: Generative AI and Collaboration Tools
 - Pilot launch of a Generative AI Agent for contextual and conversational data interaction
 - Integration of predictive models for well performance forecasting
 - Enhancement of collaboration features such as annotations, shared views, and team-based insights
4. Phase 4: Enterprise Scalability and Workflow Integration
 - Scaling the solution to cover over 10,000+ wells across the organization
 - Deployment of advanced AI agent capabilities for deeper operational insights
 - Integration with core enterprise platforms for seamless workflow alignment

Change Management and User Adoption

To ensure successful adoption of the AI-powered Digital Well Book, operator implemented a comprehensive management strategy focused on user engagement, enablement, and seamless integration into existing upstream workflows. Role-specific training programs were developed for engineers, analysts, and managers, ensuring that each user group could effectively utilize the platform's capabilities. Interactive workshops were held to demonstrate features, gather feedback, and foster hands-on experience. A super-user program was launched to build internal champions and support networks across teams.

Clear communication of the platform's benefits—such as reduced time spent on data retrieval, improved decision quality, and enhanced collaboration—was prioritized to drive user buy-in. Continuous feedback loops ensured that user needs were addressed promptly, with enhancements guided by real-world usage. Importantly, the Digital Well Book was embedded into existing upstream processes, including periodic well reviews and performance analysis workflows, enabling users to access contextual insights and expert knowledge directly within their operational routines.

Governance and Sustainability

To ensure the long-term success and sustainability of the AI-powered Digital Well Book, operator established a robust cross-functional governance framework. This framework brings together key stakeholders from IT, subsurface, operations, and data management to ensure alignment, accountability, and continuous improvement. Core responsibilities include data quality ownership and stewardship to maintain the integrity of integrated sources, as well as ongoing model retraining and validation to preserve AI accuracy and relevance. Content curation processes were also put in place to capture lessons learned and best practices, ensuring that expert knowledge is continuously enriched and accessible.

Crucially, the governance model is embedded within Operator's existing IT and data governance structures, enabling seamless integration with upstream workflows such as well reviews and operational planning. This ensures that the Digital Well Book remains a dynamic, living system—continuously evolving to meet operational needs, adapting to user feedback, and delivering sustained value across the organization.

Examples

Example 1: Well Event timelines. AI Digital well book allows you to see all the events in a timeline regardless of where the data is stored. It can extract events from data, present them on the event timeline, and connect the events together.

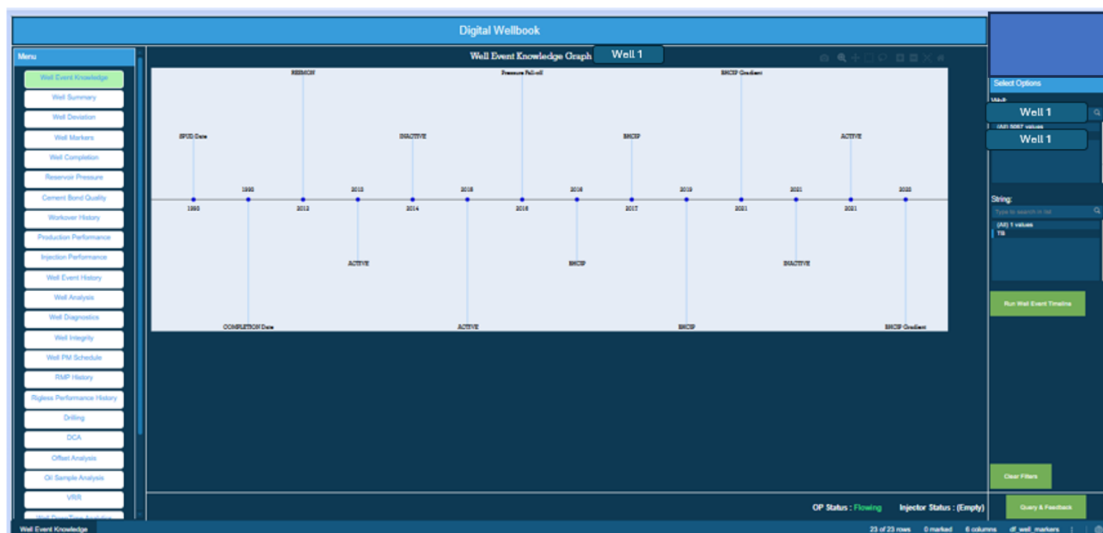


Figure 3

Example 2: Well Review elements. All well review elements are displayed in the screenshot on the left-hand page, providing comprehensive coverage from a 360-degree perspective.

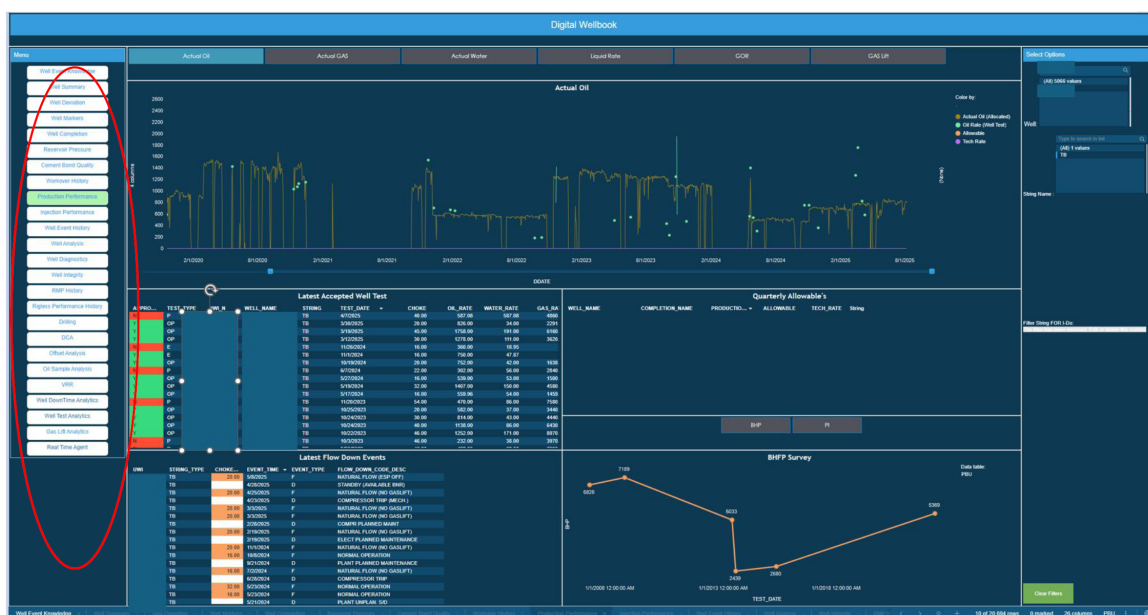


Figure 4

Example 3: MPFM Well Test – Anomaly detection. Within the AI Digital Well Book, the MPFM Well Test module leverages both historical and real-time test data to identify irregularities in multiphase flow meter (MPFM) performance. It applies machine learning models to recognize deviations from expected patterns, highlighting potential measurement errors or equipment malfunctions. By automating anomaly detection, it reduces the need for manual validation and minimizes the risk of overlooking critical issues. Engineers receive timely alerts, enabling quicker root-cause analysis and corrective actions. This ensures higher accuracy in well testing, improved reliability of production data, and more informed decision-making.

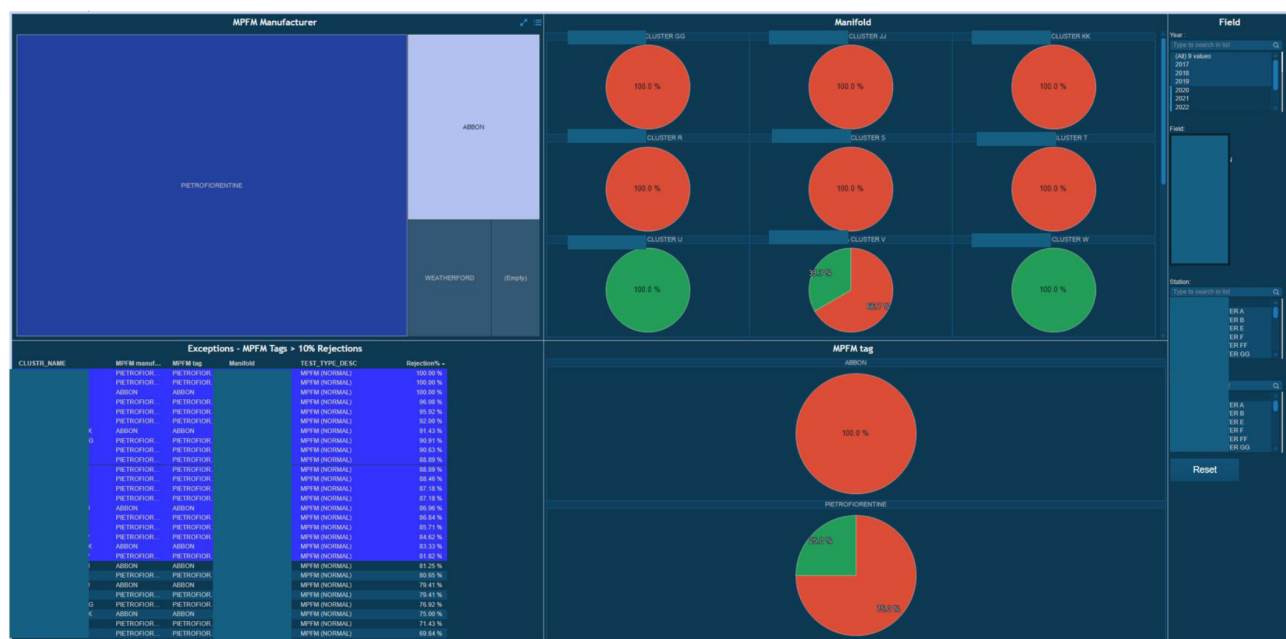


Figure 5

Example 4: Well Profitability prediction. Well profitability prediction addresses the challenge of high water cut and underperforming wells that leverage AI to forecast future well profitability and potential to simulate oil demand scenarios. The solution integrates with predictions from portfolio optimization systems, and it provides well profitability for all the inventory. It integrates high water cut prediction to flag cost-inefficient wells early. Lack of information integration and absence of profitability-based prioritization are being resolved through advanced analytics. The AI-driven approach enhances decision-making, enabling proactive well management. The goal is to boost operational efficiency, reduce costs, and maximize production value.

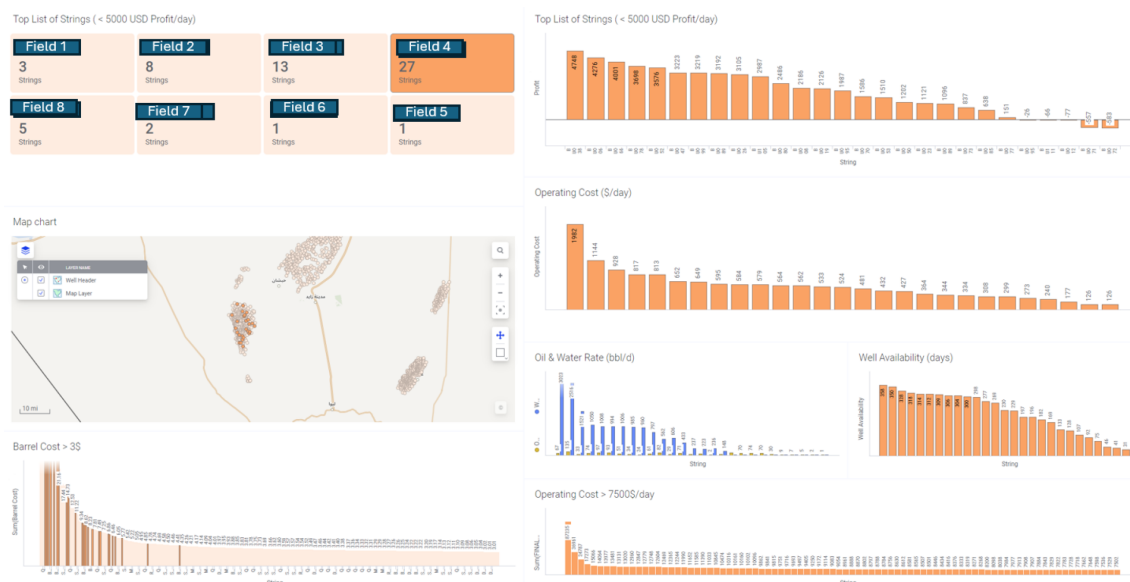


Figure 6

Example 5: The Real Time Gen AI agent. This agent acts as an intelligent conversational interface within the Digital Well Book, allowing users to access real-time well data and operational insights through natural language queries. It interprets complex, multi-part questions and maintains contextual awareness throughout the conversation (e.g., referencing previously discussed wells or operational events). Using Retrieval-Augmented Generation (RAG), it connects to the PI database to retrieve and synthesize relevant information. The agent produces concise summaries of well histories, event timelines, and performance analyses, while also identifying anomalies or correlations (e.g., pressure trends associated with ESP failures). It supports multi-modal outputs such as visual plots and timelines and ensures traceability by citing original data sources. This capability is currently under active development and construction, with ongoing enhancements to improve accuracy, safety, and integration.

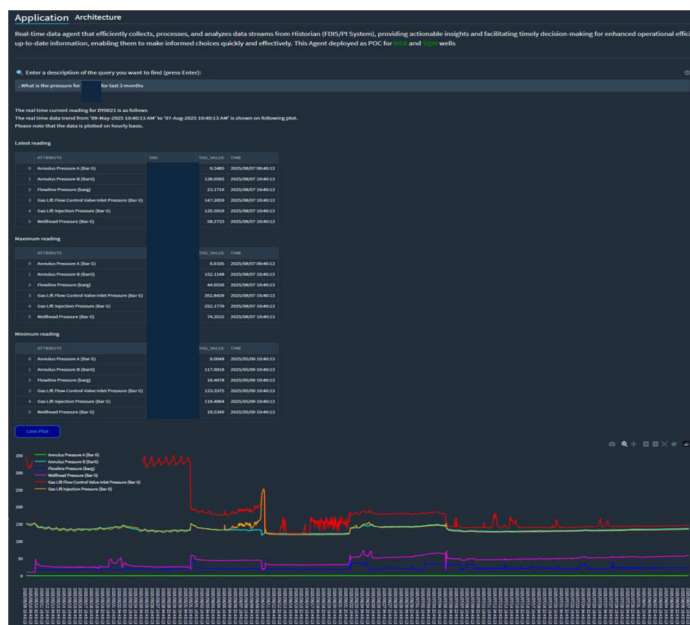


Figure 7

Benefits and Quantifiable Value

The AI Digital Well Book delivers significant, measurable value across multiple dimensions of operational efficiency. It enables a 60% reduction in data search time, allowing engineers to spend far less time hunting for information and more time on productive analysis. Well review preparation, whether for WRM or integrity assessments, is completed much faster, enabling more frequent and thorough analysis. In addition, workflows are streamlined through integrated access within key processes such as intervention planning and abandonment studies, eliminating context switching and the need for manual data consolidation.

It significantly improves decision quality by enabling fact-based choices; access to complete and contextualized historical data leads to more informed decisions on interventions, workovers, and abandonment. This comprehensive intelligence reduces risk through proactive anomaly detection and by providing critical insights from historical failures and lessons learned, thereby minimizing operational and HSE incidents. Furthermore, the system identifies key optimization opportunities, using holistic well views and AI-driven insights to reveal previously hidden possibilities for production enhancement or cost reduction.

It also drives significant improvements in knowledge management and collaboration. It preserves critical expertise by capturing and contextualizing tacit knowledge from retirees, ensuring that valuable insights are not lost. The standardized platform enhances cross-asset learning by making it easier to share best practices and lessons learned across the entire well portfolio. At the same time, it accelerates onboarding, enabling new engineers to quickly understand well history and context, thereby reducing ramp-up time and increasing their effectiveness.

In addition it delivers substantial cost savings and value creation by reducing non-productive time (NPT) through faster diagnosis and resolution of well problems. It supports optimized interventions by enabling better-informed decision-making, which increases success rates and return on investment. Additionally, it helps extend the productive life of wells by identifying opportunities to prolong output or optimize abandonment timing. Finally, by consolidating information access into a single platform, it reduces reliance on niche tools and specialized, underutilized databases, lowering licensing costs and improving overall efficiency.

Conclusion: A Foundational Step Towards the Cognitive Upstream

AI Digital Well Book represents a significant leap forward in well lifecycle knowledge management. By integrating fragmented data into a semantically rich Knowledge Graph, applying layered AI analytics, and providing intuitive access through visualization and a transformative Generative AI agent, the solution directly addresses the critical challenges of data fragmentation, expertise drain, and inefficient decision-making prevalent in the upstream industry. The measured results, notably the 60% reduction in data search time, demonstrate tangible operational efficiency gains. More importantly, it empowers engineers with unprecedented contextual understanding, enabling proactive management, optimized performance, and enhanced safety across vast well portfolios.

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Glossary

RMP – Reservoir monitoring plan
WCT – Water cut

GOR – Gas oil ratio
OPEX – Operational expenditure
IRM – Integrated Reservoir Management
MPFM – Multiphase flow meter
NF – Natural Flowing
GL – Gas-lifted
VFM – Virtual Flow Meter
PQ – Pressure Vs rate
ROI – Return on investment
WRM – Well and reservoir management

References

This manuscript is based entirely on internal knowledge, experiences, observations, and development activities. No external sources were referenced